1 **Which of the following is a scalar quantity?**

A. Electric potential
B. Force
C. Torque
D. Potential gradient

2 **If velocity of body at maximum height is half of the initial velocity, find the range of projectile fired. (initial velocity is u)**

A. $\frac{u^2}{2g}$
B. $\frac{3u^2}{2g}$
C. $\frac{\sqrt{3}u^2}{2g}$
D. $\frac{4u^2}{g}$

3 **A body is placed in an inclined plane. At certain angle, the body just starts to slide down the plane. The particular angle is _____**

A. Angle of repose
B. Angle of friction
C. Normal angle
D. Tangent angle

4 **'A' and 'B' are two bodies and A is lighter than B. If they have equal momentum, then kinetic energy of A is _____**

A. Equal to B
B. Greater than B
C. Less than B
D. Can't be predicted

5 **The area of cross section of steel wire ($Y = 2 \times 10^{10} \frac{N}{m^2}$) is $100\ m^2$. The force required to double its length is _____**

A. 2×10^{12} N
B. 2×10^{11} N
C. 2×10^{10} N
D. 2×10^{6} N

6 **Which of the following is an invert gate?**

A. NOT
B. NOR
C. OR
D. NAND

7 **Which of the following is correct about X-rays?**

A. X-ray is a longitudinal wave
B. X-ray gets diffracted by crystal
C. X-ray cannot be polarized
D. X-ray cannot ionize the gas

8 **Bragg's law has no solution if _____**

A. $\lambda > 2d$
B. $\lambda = 2d$
C. $\lambda < 2d$
D. $\lambda > d$

9 **If the rest mass of a body moving with velocity equal to 0.8 c is m_0, then the mass of body will be _____**

A. m_0
B. $2m_0$
C. $\frac{5}{3}m_0$
D. $\frac{2}{5}m_0$

10 **If the speed of rotation of earth increases, the weight of a body _____**

A. Decreases
B. May increase or decrease
C. Remains unchanged
D. Increases

11 **The increasing order of unit of energy among joule, eV, calorie and erg is _____**

A. eV, erg, Joule, Calorie
B. eV, erg, Calorie, Joule
C. erg, eV, Joule, Calorie
D. eV, Joule, erg, Calorie

12 **A compound microscope consists of an objective lens of focal length 1 cm and an eye piece of focal length 5 cm with a separation of 10 cm. The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ cm. The value of n is _____**

A. 30 cm
B. 40 cm
C. 50 cm
D. 60 cm

13 **Which of the following is equation for adiabatic process?**

A. $dQ = nC_v dt$
B. $dU = nC_p dt$
C. $dQ = nC_p dt$
D. $dU = nC_v dt$

14 **A coil of area $10\ m^2$ is placed perpendicular to a uniform magnetic field of 0.1 T, the magnetic flux linked with the coil is _____**

A. 0.1 Weber
B. 10 Weber
C. 1 Weber
D. 100 Weber

15 **A step down transformer has primary voltage 200 V and 5A current. If the ratio of turns is $\frac{1}{3}$, then secondary voltage will be _____**

A. 60V
B. 600V
C. 30V
D. 300V

16 **The elementary particle responsible for fundamental interaction is _____**

A. Fermion
B. Boson
C. Lepton
D. Hadron

17 **Nuclear fission is initiated by _____**

A. Neutron
B. Photon
C. Electron
D. Proton

18 If two temperature scales are dipped in a hot bath and reading in Fahrenheit scale is 3 times the Celsius scale. The temperature of hot bath is _____

A. 100°C
B. $\frac{80}{3}^{\circ}C$
C. 50°C
D. $\frac{50}{3}^{\circ}C$

19 If the coefficient of linear expansion of glass with mass 50 gm, the mass of liquid it holds at 80°C is _____

A. 46 gm
B. 48 gm
C. 51 gm
D. 54 gm

20 A steel tape gives correct measurement at 50°C. A piece of wood is being measured with the steel tape at 0°C, the reading is 50 cm on tape. The real length of piece of wood must be _____

A. 50 cm
B. > 50 cm
C. < 50 cm
D. Can't be predicted

21 In which condition transistor works on saturation mode?

A. V_B greater than V_E and V_C
B. V_E greater than V_B and V_C
C. V_B less than V_E and V_C
D. V_C greater than V_B and V_E

22 The focal length of convex lens depends upon _____

A. λ of incident ray
B. λ & f of incident ray
C. f of incident ray
D. Neither on λ nor on frequency of incident wavelength

23 Two thin lenses of focal length f_1 and f_2 are placed in contact co-axially. The combination is equivalent to a single lens of power _____

A. $\frac{f_1 f_2}{f_1 + f_2}$
B. $f_1 - f_2$
C. $f_1 + f_2$
D. $\frac{f_1 + f_2}{f_1 f_2}$

24 If the work function of a metal is 4eV and the velocity of ejected electron is zero. Then find the wavelength of incoming incident photon.

A. 2100 Å
B. 2300 Å
C. 3100 Å
D. 2700 Å

25 The PN junction will be forward biased if _____

A. Both P and N junction are grounded.
B. P-region is grounded and N-region is given +ve charge.
C. P-region is given positive charge and N-region is grounded.
D. Both are given positive charge.

26 If force constant of spring is $100\ \frac{N}{m}$, the PE of the spring is 0.5J, then extension produced is _____

A. 10 cm
B. 1 cm
C. 0.1 cm
D. 2 cm

27 A man heard thunder 6 seconds later after he saw a lightning. The temperature of air is 27°C. How far was the flash of light from the man? [Velocity of sound in air at 0°C is $332\ \frac{m}{s}$]

A. 1822 m
B. 2332 m
C. 2088 m
D. 2445 m

28 At constant temperature, by how much pressure should be increased in order to decrease volume by 10%?

A. 9.5%
B. 11.11%
C. 10%
D. 12%

29 Red and blue light are deviated by angle 10° and 12° by one prism and 8° and 10° by another prism then their ratio of dispersive power is _____

A. $\frac{9}{11}$
B. $\frac{11}{9}$
C. $\frac{5}{6}$
D. $\frac{6}{5}$

30 Ratio of intensities in consecutive maxima in a diffraction due to a single slit is _____

A. 1 : 2 : 3
B. 1 : 4 : 9
C. $1 : \frac{1}{\pi} : \frac{9}{\pi}$
D. $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$

31 Two charges, each of charge $1\ \mu C$ are separated by a distance 1 m. Then electrostatic force between the two charge is _____

A. 9×10^9 N
B. 9×10^3 N
C. 9×10^{-3} N
D. 9×10^{-6} N

32 Two bodies of masses m_1 and m_2 move in circle of radii r_1 and r_2 respectively. If they complete the circle in equal time, then the ratio of their angular speed $\frac{\omega_1}{\omega_2}$ is _____

A. $\frac{m_1}{m_2}$
B. $\frac{r_1}{r_2}$
C. $\frac{m_1 r_1}{m_2 r_2}$
D. 1

33 The capacitance of a capacitor is independent of _____

A. Quality of plates
B. Size of plates
C. Distance between plates
D. Medium between plates

34 An object is moving towards a plane mirror with a velocity (v) making a certain angle θ with normal of plane mirror. The velocity of image w.r.t. object is _____

A. $v\cos\theta$
B. $v\sin\theta$
C. $2v\cos\theta$
D. $2v\sin\theta$

35 In a capacitive circuit _____

A. V leads I by $\frac{\pi}{2}$
B. V & I are in same phase
C. I leads V by π
D. I leads V by $\frac{\pi}{2}$

36 If a galvanometer of $I_g = 1A$ and $G = 0.81\Omega$ is to be made to measure 10A current, then resistance of required shunt will be _____

A. 0.9Ω
B. 0.09Ω
C. 9Ω
D. 90Ω

37 The resistance of material at 20°C is 1.72Ω then resistance becomes ... at 100°C. [$\alpha = 0.0039K^{-1}$]

A. 1.58Ω
B. 2.62Ω
C. 22.6Ω
D. 15.8Ω

38 The scientific reason behind placing handle of a door close to the edge of a door is _____

A. To produce large torque from small effort
B. To produce larger momentum
C. To produce large force from small torque
D. To protect the door from unusual bending

39 A length of a wire is stretched by 0.1%, then resistance change will be _____

A. 0.2%
B. 0.8%
C. 0.3%
D. 0.4%

40 A slab of thickness 't' (t < d) of dielectric constant 5 is introduced between the plates of $10\ \mu F$ capacitor then, the new capacitance will be _____

A. $50\ \mu F$
B. $200\ \mu F$
C. $100\ \mu F$
D. $400\ \mu F$

41 Radius of conductor is increased by 2 times then specific resistance will be _____

A. 2 times
B. 4 times
C. Unchanged
D. 8 times

42 In a radioactive disintegration, if mother nuclei ${}_ZX^A$ undergoes β^+ decay, then daughter nuclei will be _____

A. ${}_{Z+1}Y^A$
B. ${}_{Z-1}Y^A$
C. ${}_ZY^{A+1}$
D. ${}_ZY^{A-1}$

43 If the kinetic energy of particle is 16 times of initial kinetic energy, then percentage change in de Broglie wavelength will be _____

A. 25%
B. 50%
C. 75%
D. 40%

44 If specific charge of a particle is $1.7 \times 10^{11}\ \frac{C}{kg}$, then potential difference per unit distance is $80\ \frac{V}{m}$, then acceleration will be _____

A. $14.4 \times 10^{12}\ \frac{m}{s^2}$
B. $144 \times 10^{16}\ \frac{m}{s^2}$
C. $14.4 \times 10^{16}\ \frac{m}{s^2}$
D. $144 \times 10^{12}\ \frac{m}{s^2}$

45 Zener diode conducts when _____

A. Reverse breakdown voltage of Zener diode is greater than ordinary diode
B. Reverse breakdown voltage of Zener diode is equal to ordinary diode
C. Zener diode is less doped
D. Reverse breakdown voltage of Zener diode is less than ordinary diode

46 Frequency of radio wave is 15 MHz, its wavelength is _____

A. 20 m
B. 30 m
C. 40 m
D. 15 m

47 When moving an electric charge of +5C from A to B at certain potential, 10 Joule work is done. Then find the potential difference _____

A. 0.5 V
B. 4 V
C. 5 V
D. 2 V

48 A circular disc of Radius 'R' has a hole of radius 'r' is floating in surface of water, what is the surface tension force acting on the disc?

A. $F = S \times 2\pi(R + r)$
B. $F = S \times 2\pi(R - r)$
C. $F = 2\pi r$
D. $F = S + R\pi r$

49 Two current carrying conductor carrying current in opposite direction _____

A. Attract
B. Repel
C. First attract then repel
D. First repel then attract

50 Current of 12V battery having 3ohm resistance and 4 ohm inductance initially at t = 0 is _____

A. 0 A
B. 1.2 A
C. 2.4 A
D. 3 A

51 Nylon 6, 6 is formed by the condensation of the monomers _____

A. Butane-1, 3-diene and 4-Propylene
B. Ethane-1, 2-diol and 4-Benzenedioic acid
C. Phenol and Aldehyde
D. 1,6-Diamino hexane and 1, 6 Hexanedioic acid

52 **Why is sodium extract boiled with HNO_3 before Lassaigne's test?**

A. To remove CN^- and S^{--}
B. To make the solution acidic
C. To increase basicity
D. To make solution lighter

53 **Metal used in Ziegler-Natta catalyst is _____**

A. Ti
B. Cd
C. Fe
D. Br

54 **Which of the following is not true about CO?**

A. It violates octet rule.
B. It makes rain water acidic.
C. It is pale blue in colour.
D. It is poisonous gas

55 **$C_6H_5N_2Cl + CuCl$ gives _____**

A. C_6H_6
B. $C_6H_5NH_2$
C. C_6H_5Cl
D. $C_6H_5 - C_6H_5$

56 **Chloretone is formed when chloroform reacts with _____**

A. Acetone
B. Acetaldehyde
C. Nitric acid
D. Phenol

57 **Dehydration of ethanol (C_2H_5OH) at 170°C forms _____**

A. Ethene
B. Carbanion
C. Carbene
D. Hydrogen ethyl sulphate

58 **Which of the following is amorphous form of carbon?**

A. Diamond
B. Graphite
C. Fullerene
D. Anthracite

59 **Which of the following compound does not show Lassaigne's test for nitrogen?**

A. Urea
B. Hydrazine
C. Azobenzene
D. Phenylhydrazine

60 **IUPAC name of $CH_3COCH_2CH_2CH_3$ is _____**

A. Pentan-2-one
B. Propylmethyl ketone
C. 4-Pentanone
D. Ethyl acetate

61 **Which of the following is refined by zone refining process?**

A. Si
B. V
C. Ti
D. C

62 **All of them are extracted from pyrometallurgy except _____**

A. Ca
B. Hg
C. Fe
D. Pb

63 **On ozonolysis, benzene gives _____**

A. Only ethanol
B. Ethanal
C. Methanal
D. Ethenedial

64 **Which of the following gives Cannizzaro reaction?**

A. C_6H_5CHO
B. CH_3CHO
C. CH_3CH_2CHO
D. $CH_3 - \overset{O}{\overset{\|}{C}} - CH_3$

65 **Nitrobenzene is _____**

A. m-directing
B. o-directing
C. p-directing
D. both o-and p-directing

66 **Haloarenes are less reactive than haloalkanes in nucleophilic substitution reaction. This is due to _____**

A. inductive effect
B. metamerism
C. resonance
D. tautomerism

67 **Commercially available concentrated HCl contains 38% HCl by mass. Calculate the molarity of the solution. (density $= 1.19\ \frac{gm}{ml}$)**

A. 11.38
B. 11.48
C. 12.38
D. 12.48

68 **Acetone on reaction with conc. H_2SO_4 gives _____**

A. Xylene
B. Mesitylene
C. Toluene
D. Benzene

69 **Heterolytic fission gives _____**

A. Carbocation
B. Ions
C. Free radicals
D. Carbocation & carbanion

70 **Which of the following reacts with Benedict's reagent?**

A. CH_3CHO
B. $CH_3 - \overset{O}{\overset{\|}{C}} - CH_3$
C. C_6H_5CHO
D. $R - COOH$

71 **The reaction is spontaneous if _____**

A. $\Delta H - T\Delta S$ is negative
B. $\Delta H - T\Delta S$ is positive
C. $\Delta H = T\Delta S$
D. $\Delta H - T\Delta S = 0$

72 **Calculate the pH of buffer solution containing 0.02 moles of NH_4Cl and 0.2 moles of NH_4OH per litre. (Given $pK_b = 5$)**

A. 9
B. 10
C. 11
D. 12

73 **In the reaction $CH_3COOH + NH_3 \rightleftharpoons CH_3COO^- + NH_4^+$, which acts as Bronsted acid?**

A. CH_3COO^-
B. CH_3COO^- and NH_4^+
C. CH_3COOH and NH_4^+
D. NH_3 and CH_3COOH

74 **Which of the following is the largest ion?**

A. Mg^{++} B. O^{--}
C. F^{-} D. Na^{+}

75 **If a gas X diffuses 4 times slower than hydrogen, then the molecular weight of X is _____**

A. 32 B. 64
C. 16 D. 8

76 **Ratio of vapor density of gas A and gas B is 4. If molecular weight of A is M, molecular weight of B is _____**

A. M B. $\frac{M}{4}$
C. 4M D. $\frac{M}{2}$

77 **Percentage of oxygen in NaOH is _____**

A. 16 B. 32
C. 10 D. 40

78 **In Down's process, which of the following takes place at cathode?**

A. Na^{+} gets oxidized B. Na^{+} get reduced
C. H_2 is released D. Cl_2 is released

79 **Which of the following doesn't produce Cl^{-} when reacted with $AgNO_3$ at 25°C?**

A. $CoCl_3 \cdot 6NH_3$ B. $CoCl_3 \cdot 3NH_3$
C. $CoCl_3 \cdot 5NH_3$ D. $CoCl_3 \cdot 4NH_3$

80 **Four metal A, B, C and D have standard reduction electric potential as $-3V$, $-2V$, 0V and 1V. The metal that will be most reducing is _____**

A. A B. B
C. C D. D

81 **Zn can give H_2 gas with HCl and H_2SO_4 but not with HNO_3, why?**

A. Zn acts as an oxidizing agent when reacted with HNO_3
B. HNO_3 is weaker acid than H_2SO_4 & HCl
C. Zn is above H_2 in electrochemical series
D. NO_3^{-} is reduced in preference to hydronium ion

82 **All of the following are true for nitric acid except _____**

A. It is colorless
B. It is powerful reducing agent & reduces noble gas
C. Its product depends upon its concentration when reacted with metals
D. It exists as planar molecule

83 **Which of the following acts both as oxidizing agent and reducing agent?**

A. HNO_3 B. HNO_2
C. H_2S D. H_2SO_4

84 **The half life of a reaction is constant on doubling the concentration of reactant, the order of reaction is _____**

A. 0 B. 1
C. 2 D. 3

85 **If the unit of rate constant of a reaction is same as that of unit of rate of reaction, the order of reaction is _____**

A. 0 B. 1
C. 2 D. 3

86 **When the concentration of reactant is doubled, then equilibrium constant becomes _____**

A. Doubled B. Halved
C. Quadrupled D. Constant

87 **If $K_c = 2$, which of the following is true?**

A. conc. of product and reactant remain same
B. conc. of product is twice than reactant
C. conc. of reactant is twice of product
D. conc. of product is half of reactant

88 $SO_2 + \frac{1}{2}O_2 \rightarrow SO_3 + \text{heat}$**, All of the following favors formation of SO_3 except _____**

A. Increase in concentration of reactants
B. Increase in pressure
C. Decrease in temperature
D. Increase in temperature

89 **How many moles of $KMnO_4$ are required to oxidize 1 mole of Sn^{+2} in acidic medium?**

A. $\frac{1}{5}$ B. $\frac{2}{5}$
C. $\frac{3}{5}$ D. $\frac{4}{5}$

90 **If $C + O_2 \rightarrow CO_2$; $\Delta H = -a$ and $2CO + O_2 \rightarrow 2CO_2$; $\Delta H = -b$. Find the heat of formation of CO.**

A. $\frac{b-2a}{2}$ B. $\frac{b-a}{2}$
C. $\frac{a-2b}{2}$ D. $a - b$

91 **Transition metals show variable oxidation state due to emission of electron from _____**

A. (n - 1)d and ns orbital B. (n - 1)d orbital
C. ns orbital D. np orbital

92 **Which of following is used as both fertilizer and explosive?**

A. Ammonium nitrate B. Urea
C. Calcium Ammonium Nitrate D. Super phosphate of lime

93 **Which of the following have/bears color to the dye?**

A. Chromophore
B. Cytochrome
C. Chromogens
D. Auxochrome

94 **Aldehyde is prepared from acid chloride by which of the following reactions?**

A. Rosenmund reaction
B. Cannizzaro reaction
C. Benzoin condensation
D. Perkin's condensation

95 **Galvanization is a process of coating zinc over iron, during this process, Zn acts as _____**

A. Sacrificial anode
B. Sacrificial cathode
C. Anode mud
D. Both b and c

96 **Which of the following produces Cl_2 at room temperature?**

A. $HCl + KMnO_4$
B. $NaCl + MnO_2$
C. $HBr + KCl$
D. $Cl_2 + NaOH$

97 **Ionization energy is minimum for _____**

A. $1s^2 2s^2 2p^6$
B. $1s^2 2s^2 2p^5$
C. $1s^2 2s^2 2p^6 3s^1$
D. $1s^2 2s^2 2p^3$

98 **Halogen with highest electron affinity is _____**

A. Cl_2
B. Br_2
C. I_2
D. F_2

99 **The electronic configuration of M^{+2} is 2, 8, 14. The mass number is 56. the no. of neutron is _____**

A. 30
B. 32
C. 31
D. 28

100 **In protein, amino acid are joined by _____**

A. Peptide bond
B. Glycosidic bond
C. Phosphodiester bond
D. Hydrogen bond

101 **Cystic fibrosis is _____**

A. Auto immune disease
B. Genetic disorder
C. Communicable disease
D. Non-communicable disease

102 **Person suffering from Sickle cell anemia is _____**

A. More prone to Typhoid
B. More prone to Malaria
C. Less prone to Typhoid
D. Less prone to Malaria

103 **All are true for cruciferae except _____**

A. Didynamous
B. False septum
C. Six stamens
D. Tetradynamous

104 **All of the following are aneuploidy except _____**

A. Turner's syndrome
B. Down's syndrome
C. Phenylketonuria
D. Klinefelter's syndrome

105 **At meiosis, centromere separation occurs at _____**

A. Prophase II
B. Anaphase II
C. Metaphase II
D. Telophase II

106 **Which of the following is diploid spore?**

A. Zygospore
B. Zoospore
C. Ascospore
D. Basidiospore

107 **The possible blood group of offspring if father has blood group A and mother has blood group B _____**

A. A, AB
B. B, AB
C. A, B, O
D. A, B, AB, O

108 **Which of the following is produced during photophosphorylation?**

A. ATP
B. ATP and $NADPH_2$
C. ATP, H_2O
D. H_2O and O_2

109 **DNA is made up of _____**

A. Pentose sugar + Nitrogen base
B. Nucleoside
C. Nucleoside + Sugar
D. Nucleoside + Phosphate

110 **Cucurbits have _____**

A. Collateral vascular bundle
B. Bicollateral vascular bundle
C. Radial vascular bundle
D. Concentric vascular bundle

111 **Both are benefitted by each other and can't live without one _____**

A. Proto-cooperation
B. Mutualism
C. Commensalism
D. Antibiosis

112 **In DNA if cytosine is 10%, amount of thymine is _____**

A. 10%
B. 20%
C. 40%
D. 50%

113 **Velamen tissue are found in _____**

A. Xerophyte
B. Epiphyte
C. Hydrophyte
D. Mesophyte

114 **Pyramid of energy is _____**

A. Spindle shape
B. Always erect
C. Inverted
D. Sometimes erect

115 **Enzyme that cuts DNA at specific site is _____**

A. Topoisomerase
B. Helicase
C. Unwindase
D. Restriction endonuclease

116 **Grafting is not possible in monocot due to _____**

A. Absence of cambium
B. Scattered vascular bundle
C. Smaller stem
D. Lack of vascular tissue

117 **Which one is used in culture medium?**

A. Laminaria
B. Gelidium
C. Alginic acid
D. Spirulina

118 **Which of the following is disaccharide?**

A. Maltose
B. Glucose
C. Cellulose
D. Starch

119 **In maize male anther matures faster than female stigma; maize is _____**

A. Heterostyly
B. Herkogamy
C. Protandry
D. Protogyny

120 **Gemma cup is found in _____**

A. Cycas
B. Funaria
C. Mosses
D. Marchantia

121 **A colorblind man marries a normal woman whose father was colorblind, then the probability of son to be colorblind is _____**

A. 25%
B. 50%
C. 75%
D. 100%

122 **In photosynthesis, O_2 is evolved from water during _____**

A. Oxidative photophosphorylation
B. Photolysis of water
C. Excitation of electron
D. Excitation of cell

123 **Arrangement of cilia and flagella in eukaryotes _____**

A. 9 + 2
B. 9 + 0
C. 9 + 1
D. 9 + 3

124 **Dissolved oxygen is decreased in pond due to _____**

A. Biomagnification
B. Eutrophication
C. Nitrification
D. Denitrification

125 **When pyruvic acid is changed to acetyl CoA, it is _____**

A. Oxidized
B. Reduced
C. Isomerized
D. Rearrangement

126 **Which of the following has narrow lumen?**

A. Fibres
B. Sclereids
C. Tracheids
D. Vessels

127 **Blocking of sieve tube of phloem is due to _____**

A. Tyloses
B. Callose
C. Aerenchyma
D. Collenchyma

128 **If number of chromosome in Pisum sativum is 2n = 14, then the number of linkage group is _____**

A. 14
B. 7
C. 8
D. 15

129 **Chromosome with centromere at center _____**

A. Metacentric
B. Sub-metacentric
C. Telocentric
D. Acrocentric

130 **The contents of cell is expelled out through the cell membrane, the process is called _____**

A. Phagocytosis
B. Pinocytosis
C. Endocytosis
D. Exocytosis

131 **In papilionaceous, the outer large corolla is _____**

A. Keel
B. Standard
C. Wings
D. Alae

132 **Hybrid form genetically distinct plants is not possible by traditional plant breeding method but can be obtained by _____**

A. Somatic hybridization
B. Genetic engineering
C. Inbreeding
D. Outbreeding

133 **Which of the following has same genotypic and phenotypic ratio?**

A. Co-dominance
B. Complete dominance
C. Epistasis
D. Dihybrid cross

134 **A bacteria divides every 35 minutes. If a culture containing 10^5 cells per mL is grown for 175 min, what will be the cell concentration per ml after 175 minutes?**

A. 32×10^5 cells
B. 175×10^5 cells
C. 64×10^5 cells
D. 16×10^5 cells

135 **If a cross is made between diploid (2n) female and tetraploid (4n) male, the resultant endosperm will be _____**

A. 2n
B. 3n
C. 4n
D. 5n

136 **Compounds which help in metabolic reactions without itself being changed is _____**

A. Enzyme
B. Hormone
C. Nucleic acid
D. Nucleoprotein

137 **Chemical causing chromosomal abnormality is called _____**

A. Mutagen
B. Mutation
C. Mutant
D. Muton

138 **Which of the following unwinds DNA molecule?**

A. Helicase
B. Primase
C. Folic acid
D. Ligase

139 **Animals found at bottom of the sea are called _____**

A. Benthos
B. Cryptosoma
C. Phytoplankton
D. Littoral

140 **Choose the correct succession type from the given options considering the sequence of different stage as below:** Crustose lichen → Foliar lichen → Moss → Herbs → Shrub → Mesophytic tree (climax)

A. Hydrosere
B. Psammosere
C. Xerosere
D. Halosere

141 **The increasing order of complexity in canal system of Porifera is _____**

A. Ascon, Sycon, Leucon
B. Ascon, Leucon, Sycon
C. Sycon, Leucon, Ascon
D. Leucon, Sycon, Ascon

142 **Amphids present in ventrolateral lips of Ascaris is _____**

A. Chemoreceptor
B. Gustatoreceptor
C. Olfactoreceptor
D. Tactoreceptor

143 **An infected female Anopheles mosquito can infect a healthy man when _____**

A. Gametogony is completed
B. Sporogony is completed
C. Erythrocytic cycle is completed
D. Erythrocytic cycle is just started

144 **Echinodermata are brainless, heartless and headless; yet from evolutionary point of view, they have been placed on the top of invertebrates because they _____**

A. have great power of regeneration
B. have exclusively marine habit
C. are enterocoelomate
D. ammonotelic

145 **Which of the following is vertebrate?**

A. Cuttle fish
B. Globe fish
C. Cray fish
D. Devil fish

146 **Intraocular pressure in eye is measured by _____**

A. Tonometer
B. Barometer
C. Sphygmomanometer
D. Spirometer

147 **Apes are dissimilar from human due to _____**

A. Menstruation in female
B. Absence of prominent chin
C. Smaller lumbar region
D. Absence of tail

148 **If the concentration of urea is less than normal in amount of a person, this can be due to _____**

A. He is on longer hunger strike
B. Renal calculi
C. Deficiency of ADH
D. Bright disease

149 **Recapitulation theory, "Ontogeny repeats phylogeny" was proposed by _____**

A. Haeckel
B. Hugo de Vries
C. Darwin
D. Lamarck

150 **Grinding organ of Earthworm is _____**

A. Pharynx
B. Gizzard
C. Stomach
D. Intestine

151 **Intestinal caeca is present in _____**

A. 30th segment
B. 14th, 15th &16th segments
C. 27th segment
D. 26th segment

152 **Which of the following gland is only present in reptiles?**

A. Mucous gland
B. Femoral gland
C. Sebaceous gland
D. Uropygial gland

153 **All are true except _____**

A. Blue whale: Ammonotelic
B. Lizard: Uricotelic
C. Human: Ureotelic
D. Aves: Uricotelic

154 **A boy sustained an accident and needs emergency blood and doesn't know the blood group. A doctor's friend after consulting the doctor donated him blood immediately. The blood group doctor's friend must have is _____**

A. O^-
B. AB^+
C. O^+
D. AB^-

155 **If the fallopian is blocked, which of the following method is used to produce baby?**

A. GIFT
B. IVF
C. VIFT
D. SVT

156 **4S gland _____**

A. Pancreas
B. Pituitary
C. Hypothalamus
D. Adrenal gland

157 **Areolar tissue connects _____**

A. Bone to bone
B. Muscle to bone
C. Integument and muscles
D. Muscle to fat

158 **Which one carries impulse towards CNS?**

A. Auditory
B. Abducens
C. Trochlear
D. Ventral root horn of spinal cord

159 **If the mastication area is paralyzed, which of the following nerves is affected?**

A. Optic
B. Trigeminal
C. Oculomotor
D. Vagus

160 **Which of the following cells is not a part of CNS?**

A. Microglial cell
B. Schwann cell
C. Ependymal cell
D. Astrocyte cell

161 **A pap smear is done around vagina and cervix; what kind of tissue may line those area?**

A. Simple cuboidal
B. Keratinized squamous epithelium
C. Squamous epithelium
D. Non-keratinized stratified squamous epithelium

162 **Thyroid gland is lined by _____**

A. Simple cuboidal epithelium
B. Simple columnar epithelium
C. Squamous epithelium
D. Pseudostratified columnar epithelium

163 **The stomach of frog attached to the dorsal wall by means of mesentery called _____**

A. Oxygaster
B. Endogaster
C. Mesogaster
D. Megagaster

164 **Mucin present in saliva is _____**

A. Glycoprotein
B. Lipoprotein
C. Phosphoprotein
D. Chromoprotein

165 **Which of the following sets of disease is viral disease?**

A. Typhoid, TB
B. AIDS, Kala-azar
C. Common cold, Herpes
D. Pneumonia and TB

166 **Body cavity of Echinodermata is formed by _____**

A. Splitting of mesodermal band
B. Arrangement of mesenchymal cells
C. Persistent blastocoel
D. Archenteron pouch in larval stage

167 **Specific antibodies are extracted for the diagnosis of cancer in early days. These antibodies are called _____**

A. Monoclonal antibody
B. Polyclonal antibody
C. Myelonal antibody
D. Biclonal antibody

168 **Enterokinase differs from other intestinal juice like Erepsin, Maltase, Lactase, etc as _____**

A. It is an enzyme
B. It is an intestinal juice
C. It needs to be activated
D. It does not digest food

169 **Digestion of protein gives _____**

A. Glucose
B. Amino acid
C. Maltose
D. Glycerol

170 **If there is deficiency of iodine, then metamorphosis of tadpole of frog _____**

A. Delays
B. Stops
C. Accelerates
D. No change

171 **All are true for dengue and malaria except _____**

A. Both are viral diseases
B. Sudden fever
C. Both are caused by bite of mosquito
D. Muscular pain

172 **Muscle co-ordination and control is done by _____**

A. Cerebrum
B. Cerebellum
C. Pons
D. Medulla oblongata

173 **$Na^+ - K^+$ ATPase pump pumps 3Na^+ outside the axoplasm and K^+ inside the axolemma. If 3Na^+ atom is pumped out, but only one K^+ atom is pumped inside, the axoplasm would be _____**

A. More positive
B. More negative
C. Neutral
D. Same charge

174 **Human embryo has pharyngeal gill slits and tail. This represents that _____**

A. Mammals have evolved from fish like ancestor
B. Fishes and mammals are not related
C. Both fish and mammal have common ancestor
D. Fishes have evolved from mammals

175 **Darwin believed that certain part of body become more larger and complex through generations because they _____**

A. are used more extensively than other parts
B. contribute to greater reproductive success
C. are predetermined to do so
D. are most similar to god's perfections

176 **Although aves have small lungs, they can respire effectively as exchange of gases takes place in _____**

A. Air sacs
B. Alveoli
C. Open capillaries
D. Blood

177 **All are the function of Na^+ except _____**

A. It binds with glucose
B. Produces protein and enzyme
C. Nerve impulse conduction
D. Regulates heart activity

178 **If earthworm is pricked by needle, then which fluid comes out of its body?**

A. Coelomic fluid
B. Blood
C. Lymph
D. Plasma

179 **Highly concentrated urine is formed _____**

A. Low ADH secretion
B. Polyuria
C. By drinking a lot of water
D. More number of Juxtamedullary nephrons

180 **You are going on a Shivapuri hike with few of your friends. What type of food would you prefer for sustain/persistent source of energy?**

A. Complex carbohydrate from whole grain
B. Sugar containing chocolate, bar
C. Sporty drinks
D. Fiber containing carbohydrate

181 **Find the nth term of the following sequence: 5, 9, 13, 17, ...**

A. $4n + 1$
B. $2n$
C. $2n + 1$
D. $4n - 2$

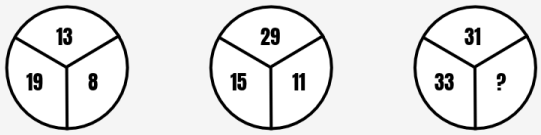
182 **The average of five consecutive numbers A, B, C, D & E is 48. Then, the product of A & E is _____**

A. 2400
B. 2350
C. 2300
D. 2332

183 **Rashmi set the time in a clock correctly at 10AM. In every hour, the clock gains 12 minutes. Find out the correct time if the clock shows 1 PM?**

A. 11:30 am
B. 12:15 pm
C. 12:30 pm
D. 1:30 pm

184 **Find the missing number.**



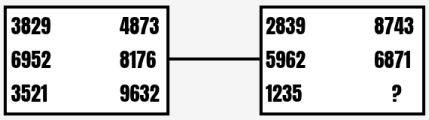
A. 6
B. 12
C. 14
D. 16

185 **Choose the appropriate option of fill in the blank given below _____**



A. 4
B. 6
C. 7
D. 9

186 **Find the missing number.**



A. 2639
B. 3971
C. 3212
D. 2760

187 **Eight friends are sitting around a round table at equal angles. Bikesh is facing south. Shyam and Sajan are at the left and right of Bikesh respectively. Pooja is not adjacent to Ram who is facing east. Sita is third to the right of Hari. Kamal is opposite to Sajan. Then who is opposite to Bikesh?**

A. Shyam
B. Hari
C. Pooja
D. Kamal

188 **A has one and a half more as B. B has one and a half more than C. The total amount of gold is 76 ounces. Then find, how many gold in ounces does these individuals have each respectively?**

A. 38, 22, 16
B. 36, 24, 16
C. 40, 24, 12
D. 46, 22, 18

189 **Son wants to buy a dog but his parents say: (I) It is not allowed in the quarter of office. (II) I will buy aquarium for you. What should be concluded from above statement?**

A. Parents like fish more than dog
B. They live in quarter
C. Son get the fish
D. Parents didn't like dog

190 **The mirror image of a clock is given. What time is shown on the clock?**



A. 6:30
B. 6:00
C. 5:30
D. 8:30

191 Sanjeev takes 8 hours to copy 50 pages manuscript while Sandesh can copy the same manuscript in 6 hours. How many would it take them to copy similar 100 pages manuscript, if they work together?

A. $9\frac{5}{7}$ B. $6\frac{6}{7}$

C. $10\frac{2}{3}$ D. 14

192 Select the alternatives which represents there out of the five alternative figures which when fitted into each other would form a complete square.



A. 123 B. 125

C. 234 D. 145

193 In search of his father, Jenish started moving 90m east then turned right and went 20m. After this, he again turned right, after going 30m he reached to his uncle's house but didn't meet his father. From there he started moving towards north and moved 100m & met his father. How far did he meet his father from the starting point?

A. 80 m B. 100 m

C. 140 m D. 260 m

194 Which of the following will continue the given series? $\frac{1}{3}, \frac{7}{10}, \frac{13}{17}, \frac{19}{24}, ?$

A. $\frac{25}{31}$ B. $\frac{25}{30}$

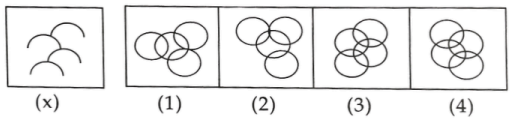
C. $\frac{26}{30}$ D. $\frac{27}{35}$

195 Artist : Brush :: Scribe : .....

A. Book B. Paper

C. Word D. Pen

196 Find out the alternative figure which contains figure (X) as its part.



A. 1 B. 2

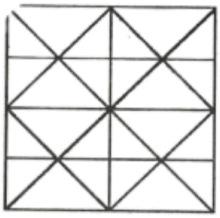
C. 3 D. 4

197 A train is travelling at the speed of 90 mph enters a tunnel 3.5 miles long. The length of the train is 0.25 miles. How long does it take for all of the train to pass throughout the tunnel, from the moment it enters until emerging out completely?

A. 2.5 min B. 2 min

C. 2.32 min D. 3 min

198 Count the number of squares in the given figure.



A. 5 B. 10

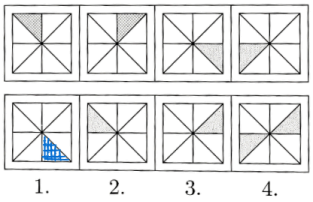
C. 12 D. 14

199 A queue of 50 students is made placing all girls at first and then followed by the boys. If Sambridhi is ranked 36th from last and she is at the bottom place from the girls then, what is the number of boys in that queue?

A. 14 B. 15

C. 35 D. 36

200 Select the figure that will continue the same series established by the given figures.



A. 4 B. 2

C. 1 D. 3